

【物理化学特論】 5. Hydrogenic atoms

No. : _____ Name : _____

- (1) Rydberg discovered that the inverse of the wavelength of the radiation from hydrogen atoms is proportional to $(1/n^2 - 1/m^2)$, where n and m are integers. Deduce the expression for the Rydberg constant R_∞ from the energy levels of hydrogenic atoms.
- (2) Calculate the lowest excitation energy for hydrogenic atoms in [eV].
- (3) Convert the energy in (2) into wavelength. Answer the name of the electromagnetic wave at the wavelength.
- (4) Express the mean radii $\langle r \rangle$ for the 1s and 2s orbitals as a function of the Bohr radius a_0 .
- (5) Express the most-probable radius (maximum in the radial distribution function) for electron in the 1s and 2s orbitals as a function of the Bohr radius a_0 .